every time NASA describes as exploration sending a few astronauts 200 miles up in a small capsule that endlessly circles the Earth and goes nowhere. Compared to the brilliant achievements of robotic missions, it is striking how rarely fundamental scientific findings emerge from manned missions. Except for repairing ineptly manufactured or malfunctioning satellites, or launching a satellite that could just as well have been sent up in an unmanned booster, the manned program has, since the 1970s, seemed unable to generate accomplishments commensurate with the cost. Others looked at NASA as a stalking horse for grandiose schemes to put weapons into space, despite the fact that an orbiting weapon is in many circumstances a sitting duck. And NASA showed many symptoms of an aging, arteriosclerotic, overcautious, unadventurous bureaucracy. The trend is perhaps beginning to be reversed.

But these criticisms—many of them surely valid—should not blind us to NASA triumphs in the same period: the first exploration of the Uranus and Neptune systems, the in-orbit repair of the Hubble space telescope, the proof that the existence of galaxies is compatible with the Big Bang, the first close-up observations of asteroids, mapping Venus pole to pole, monitoring ozone depletion, demonstrating the existence of a black hole with the mass of a billion suns at the center of a nearby galaxy, and a historic commitment to joint space endeavors by the U.S. and Russia.

There are far-reaching, visionary, and even revolutionary implications to the space program. Communications satellites link up the planet, are central to the global economy, and, through television, routinely convey the essential fact that we live in a global community. Meteorological satellites predict the weather, save lives in hurricanes and tornados, and avoid many billions of dollars in crop losses every year. Military-reconnaissance and treaty-verification satellites make nations and the global civilization more secure; in a world with tens of thousands of nuclear weapons, they calm the hotheads and paranoids on all sides; they are essential tools for survival on a troubled and unpredictable planet.

Earth-observing satellites, especially a new generation soon to be deployed, monitor the health of the global environment: greenhouse